

total variation
 $SST = 30, df = N - 1 = 8$

between groups
 $SSB = 24, df = k - 1 = 2$

within groups
 $SSW = 6, df = N - k = 6$

$\div df$

$\div df$

$MSB = 24/2 = 12$

$MSW = 6/6 = 1$

$F = MSB/MSW = 12$